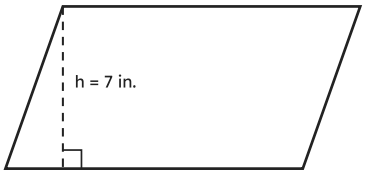
Benchmark: **MA.7.GR.1.1** · Area of trapezoids, parallelograms, and rhombi Student code: _____ Calculator: yes

Learning goal: I can find the area of trapezoids, parallelograms, and rhombi. (circle your level after scoring)

4	I can find a missing dimension when the area is given, or explain where a formula comes from.
3	I can use the formulas to find the area of a trapezoid, parallelogram, or rhombus.
2	I can find the area of a rectangle and a triangle.
1	I need help to start this benchmark.

1. The parallelogram shown has a base of 18 inches and a vertical height of 7 inches.

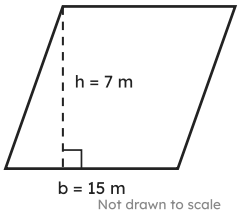
What is its area, in square inches?



Not drawn to scale

2. The rhombus shown has a base of 15 meters and a vertical height of 7 meters.

What is its area, in square meters?



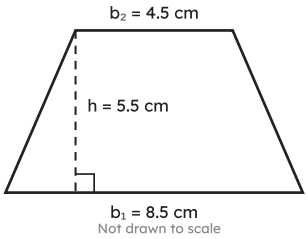
Not drawn to scale

Turn over →

Benchmark: **MA.7.GR.1.1** · Area of trapezoids, parallelograms, and rhombi Student code: _____

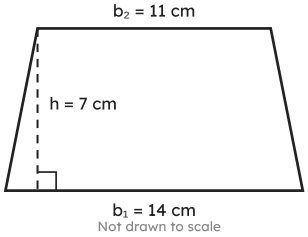
3. A trapezoid has bases of 8.5 centimeters and 4.5 centimeters and a vertical height of 5.5 centimeters.

What is its area, in square centimeters?



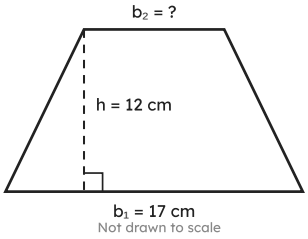
4. A trapezoid has bases of 14 cm and 11 cm and a vertical height of 7 cm.

What is its area, in square centimeters?



5. The trapezoid shown has a base of 17 centimeters and a vertical height of 12 centimeters. Its area is equal to the area of a parallelogram with a base of 15 centimeters and a vertical height of 10 centimeters.

What is the length of the trapezoid's other base, in centimeters?



Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.1.1 · MA.7.GR.1.1 · Area of trapezoids, parallelograms, and rhombi

Unit 4: Area. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $126\,\text{in.}^2$

$A = bh = 18 \times 7 = 126\,\text{in.}^2$

2. Regular $105\,\text{m}^2$

A rhombus is a parallelogram, so $A = bh$.

$A = 15 \times 7 = 105\,\text{m}^2$.

3. Regular $35.75\,\text{cm}^2$

$A = \frac{1}{2}(b_1 + b_2)h$.

$\frac{1}{2}(8.5 + 4.5)(5.5) = \frac{1}{2}(13)(5.5) = 35.75\,\text{cm}^2$.

4. Regular $87.5\,\text{cm}^2$

$A = \frac{1}{2}(b_1 + b_2)h$.

$\frac{1}{2}(14 + 11)(7) = \frac{1}{2}(25)(7) = 87.5\,\text{cm}^2$.

5. Rigor **8 cm**

Area of the parallelogram: $A = bh = 15 \times 10 = 150\,\text{cm}^2$.

The trapezoid has the same area: $A = \frac{1}{2}(b_1 + b_2)h$, so $150 = \frac{1}{2}(17 + b)(12) = 6(17 + b)$.

$17 + b = 150 \div 6 = 25$.

$b = 25 - 17 = 8\,\text{cm}$.

Re-check strip (3 items, C-GR.1.1-R)

1. Regular $153\,\text{in.}^2$

$A = bh = 17 \times 9 = 153\,\text{in.}^2$

2. Regular **C.** $180\,\text{in.}^2$

A rhombus is a parallelogram, so $A = bh$.

$18 \times 10 = 180\,\text{in.}^2$

3. Regular $99\,\text{cm}^2$

$A = \frac{1}{2}(b_1 + b_2)h$.

$\frac{1}{2}(14 + 8)(9) = \frac{1}{2}(22)(9) = 99\,\text{cm}^2$.

Comparison strip (2 items, C-GR.1.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $35\,\text{m}^2$

A rhombus is a parallelogram, so $A = bh$.

$A = 7 \times 5 = 35\,\text{m}^2$.

2. Regular $66\,\text{in.}^2$

$A = bh = 6 \times 11 = 66\,\text{in.}^2$

Reteach plan — C-GR.1.1

MA.7.GR.1.1 · Area of trapezoids, parallelograms, and rhombi

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

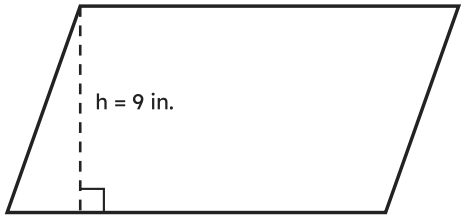
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.1.1-R

Benchmark: **MA.7.GR.1.1** · Area of trapezoids, parallelograms, and rhombi Student code: _____ Calculator: yes

1. The parallelogram shown has a base of 17 inches and a vertical height of 9 inches.

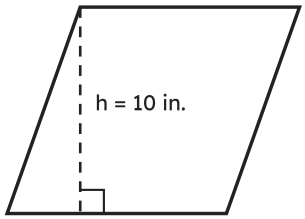
What is its area, in square inches?



b = 17 in.
Not drawn to scale

2. A rhombus has a base of 18 inches and a vertical height of 10 inches.

What is its area, in square inches?



b = 18 in.
Not drawn to scale

A. $324\,\text{in.}^2$

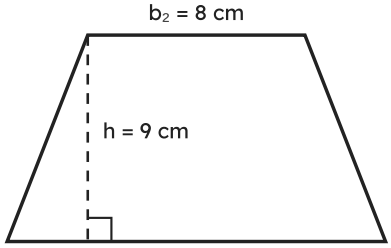
B. $90\,\text{in.}^2$

C. $180\,\text{in.}^2$

D. $72\,\text{in.}^2$

3. A trapezoid has bases of 14 cm and 8 cm and a vertical height of 9 cm.

What is its area, in square centimeters?



Not drawn to scale

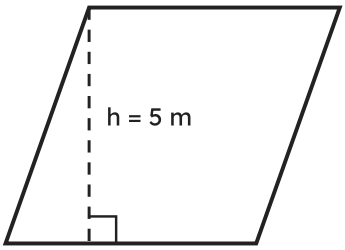
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-GR.1.1-X

Benchmark: **MA.7.GR.1.1** · Area of trapezoids, parallelograms, and rhombi Student code: _____ Calculator: yes

1. The rhombus shown has a base of 7 meters and a vertical height of 5 meters.

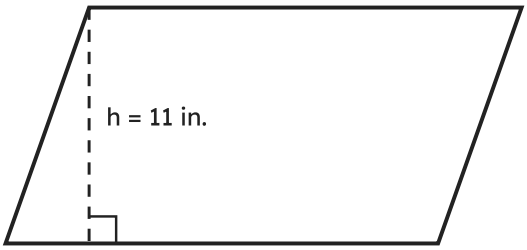
What is its area, in square meters?



Not drawn to scale

2. The parallelogram shown has a base of 6 inches and a vertical height of 11 inches.

What is its area, in square inches?



Not drawn to scale

Score: ____ / 2 → copy to your progress strip